

Project Report: Titanic Survival Prediction using kNN

Which dataset did I choose and why: For this project, I chose the Titanic Dataset. I selected this database because it has a clear target variable. It is a Classification problem, and our goal is to predict one of two possible outcomes: whether the passenger survived or not.

Preprocessing steps: Before feeding the data into the model, I performed the following steps:

- Selected the most important features (Pclass, Sex, Age, SibSp, Parch, Fare).
- Imputed the missing values in the Age and Fare columns with medians.
- Replaced the text-based column determining gender with numerical values (0 and 1).
- Since k-NN is based on distance calculations, I used StandardScaler to bring all numerical data to the same scale.

How did I select k: To find the optimal number of k neighbors, I used the GridSearchCV method. I tested all odd values of k from 1 to 31. Additionally, I built a plot that visually demonstrated at which k value the model achieves maximum accuracy on the test data.

Which distance metric worked best: The euclidean and manhattan metrics were compared using GridSearchCV. In my experiments, the euclidean option performed best because it calculated the "similarity" between passengers more accurately for these features.

Final performance results: The model demonstrated stable results on the test data, achieving 80% accuracy. This indicates that the model successfully generalized the data.

What did I learn:

- Learned that checking for missing values is critical, as the k-NN algorithm cannot work with empty values.
- Observed that k-NN can "memorize" (overfit) the training data if k is too small.
- Saw how gender and social class can be decisive factors in the prediction.

Limitations of my model: Despite the satisfactory accuracy achieved, the k-NN model has several limitations that could affect its efficiency in real-world scenarios:

- **Sensitivity to Noise:** k-NN relies on distance calculation. If there are columns in the data that are unrelated to the probability of survival and were not removed, they can confuse the model.
- **Scalability / Memory Usage:** k-NN does not build a mathematical formula; instead, it stores all data in memory. If there were 1 million rows instead of 891 Titanic rows, the model would run extremely slowly.